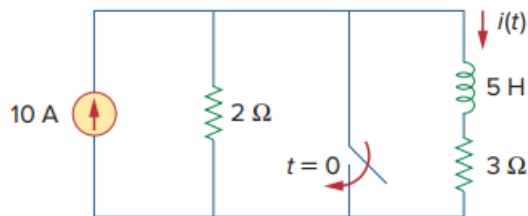


Problem

For the circuit in Fig. 7.80, the inductor current just before $t = 0$ is:

- (a) 8 A
- (b) 6 A
- (c) 4 A
- (d) 2 A
- (e) 0 A

Fig. 7.80:



Step-by-step solution

Step 1 of 2

Refer to circuit diagram in Figure 7.80 in the textbook.

At $t = 0^-$, the switch in the circuit is open for long time and circuit reaches steady state. Under steady state, inductor acts as an open circuit. The circuit diagram for $t = 0^-$ is shown in Figure 1.

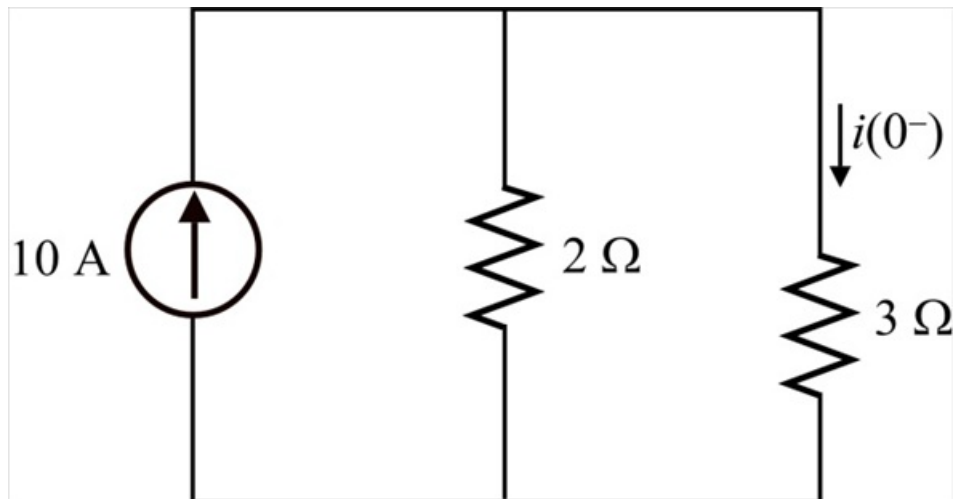


Figure 1

[Comments \(1\)](#)



Anonymous

Diagram is correct but description says open circuit instead of short circuit.

Step 2 of 2

Calculate the current through inductor at $t = 0^-$ using current division rule.

$$\begin{aligned} i(0^-) &= \left(\frac{2 \, \Omega}{2 \, \Omega + 3 \, \Omega} \right) (10 \, \text{A}) \\ &= \left(\frac{2 \, \Omega}{5 \, \Omega} \right) (10 \, \text{A}) \\ &= (0.4)(10 \, \text{A}) \\ &= 4 \, \text{A} \end{aligned}$$

The current through inductor at $t = 0^-$ is $4 \, \text{A}$.

Therefore, the correct option is c.